

The Chi-Squared Distribution

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Introduction

The chi-squared distribution arises as the sum of squared independent standard normal random variables. It is the sampling distribution of the sample variance under normality, the reference distribution for goodness-of-fit and contingency tests, and a key ingredient in the F distribution used throughout ANOVA.

Prerequisites

Normal distribution, basic calculus.

Theory

If $Z_1, \dots, Z_k$ are iid standard normal, then

$$Q = \sum_{i=1}^k Z_i^2 \sim \chi_k^2.$$

The parameter k is the **degrees of freedom**. The PDF is

$$f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}, \quad x > 0.$$

Moments: $E[Q] = k$, $\text{Var}(Q) = 2k$.

Sum property: if $Q_1 \sim \chi_{k_1}^2$ and $Q_2 \sim \chi_{k_2}^2$ are independent, then $Q_1 + Q_2 \sim \chi_{k_1+k_2}^2$.

Sample variance: for iid normal $X_1, \dots, X_n$ with variance σ^2 ,

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2.$$

This is the basis of confidence intervals for a normal variance.

Large-df behaviour: χ_k^2 is approximately normal for large k (by the CLT applied to the sum of squared normals).

Assumptions

For the sample-variance result: iid normal observations. For the goodness-of-fit approximation in chi-squared tests: independent observations and large enough expected cell counts.

R Implementation

```

k <- 5

# PDF and CDF
x <- seq(0, 20, length.out = 400)
plot(x, dchisq(x, df = k), type = "l", col = "#2A9D8F", lwd = 2,
     main = paste("Chi-squared,", k, "df"), ylab = "f(x)")

# Moments
c(theoretical_mean = k, theoretical_var = 2 * k,
  empirical_mean   = mean(rchisq(1e5, k)),
  empirical_var    = var(rchisq(1e5, k)))

# Verify: sum of k squared standard normals
n <- 1e5
Q_direct <- rchisq(n, df = k)
Q_from_normals <- rowSums(matrix(rnorm(n * k), n, k)^2)
c(mean_direct = mean(Q_direct), mean_from_normals = mean(Q_from_normals))

# Confidence interval for a variance
set.seed(2026)
X <- rnorm(30, mean = 0, sd = 2)
s2 <- var(X)
n <- length(X)
chi_l <- qchisq(0.025, n - 1); chi_u <- qchisq(0.975, n - 1)
c((n - 1) * s2 / chi_u, (n - 1) * s2 / chi_l)

```

Output & Results

```

theoretical_mean theoretical_var empirical_mean empirical_var
                5.0             10.0           5.003         9.98

```

```

mean_direct mean_from_normals
          5.005             5.004

```

```
[1] 2.542 7.206
```

A 95% CI for the true variance (which is 4) is (2.54, 7.21), which captures the true value. The sum-of-squared-normals construction produces the expected mean of 5.

Interpretation

Chi-squared tests of independence, goodness-of-fit, and variance homogeneity all reference the chi-squared distribution. Reporting a test statistic value like “ $\chi^2_3 = 9.5$, $p = 0.023$ ” is reporting a chi-squared-distributed statistic with its p-value.

Practical Tips

- Chi-squared confidence intervals for variance are highly sensitive to the normality assumption.
- For large df, use the normal approximation: $\chi^2_k \approx \mathcal{N}(k, 2k)$, or more accurately $\sqrt{2\chi^2_k} \approx \mathcal{N}(\sqrt{2k-1}, 1)$.
- Chi-squared tests with small expected counts are poorly approximated; use exact tests or Monte Carlo.
- Non-central chi-squared (with non-centrality parameter λ) governs power calculations for chi-squared tests.
- The square of a t_ν is $F_{1,\nu}$, not chi-squared; do not conflate.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots